Controllability:

if $\exists t_1 < \infty$ so that gramian $W_c(t_0, t_1)$ is <u>positive definite</u>, where

$$W_c(t_0, t_1) = \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) B(\tau)^T \Phi(t_1, \tau)^T d\tau$$

For LTI, an equivalent, but not identical, test is:

$$\text{rank}(M_c) = n$$

where $M_c = [B \; AB \; A^2B \cdots A^{n-1}B]$

and thus $\text{span}\{A^k \underline{b}_j\} = \mathbb{R}^n$, $k = 0, \ldots, n-1$, $j = 1, \ldots, m$

$\Rightarrow$

If system controllable, the state can be transferred from arbitrary initial state $\underline{x}(t_0) = \underline{x}_0$ to arbitrary final state $\underline{x}(t_1) = \underline{x}_1$.

This does <u>not</u> imply state can be "held" at $\underline{x}(t) = \underline{x}_1$ for $t \geq t_1$.

what (additional!) condition ensures this?

To <u>hold</u> state at $\underline{x}_1$ once we get there we need $\dot{\underline{x}}(t)\Big|_{\underline{x} = \underline{x}_1} = \underline{0} \quad t \geq t_1$

i.e. $\underline{0} = A(t)\underline{x}_1 + B(t)\underline{u}(t)$

$\Rightarrow$ $A(t)\underline{x}_1 = -\sum_{j=1}^{m} \underline{b}_j(t) u_j(t)$

$\Rightarrow A(t)\underline{x}_1 \in \text{span}\{\underline{b}_j(t)\}_{j=1}^{m} \quad \forall t \geq t_1$

Since $m < n$ typically, the points $\underline{x}_1$ we can hold are restricted to an (at most) m-dim <u>subspace</u> of $\mathbb{R}^n$.

For example, with $m = 1$ input, feasible $\underline{x}_1$ must satisfy

$$A(t)\underline{x}_1 \propto \underline{b}_1(t)$$

which defines a <u>line</u> in $\mathbb{R}^n$.

If we assume LTI, and non-singular A, the

$$A\underline{x}_1 = -B\underline{u}$$

$$\Rightarrow \underline{x}_1 = -(A^{-1}B)\underline{u}$$

$$\Rightarrow \underline{x}_1 \in \text{span}\{A^{-1}\underline{b}_j\}_{j=1}^{m}$$

Back to controllability:

for LTI, we know $\Phi(t,\tau) = e^{A(t-\tau)}$

then (w/ change of variable in integral) the Gramian can be written as

$$W_c(0,t) = \int_0^t e^{A\sigma} BB^T e^{A^T\sigma} d\sigma$$

and $\lim_{t\to\infty} W_c(0,t) = \int_0^\infty \{\cdots\}\, d\sigma$

lim exists if A is "Hurwitz" if all eigenvalues in LHP

and limit matrix $W_{c,ss}$ satisfies

$$AW_{c,ss} + W_{c,ss}A^T = -BB^T$$

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a Lyapunov Eq'n

recall diff eq'n for $W_c(t) = W_c(t_0,t)$

$$\dot{W}_c(t) = A(t)W_c(t) + W_c(t)A^T(t) + B(t)B^T(t)$$

$$W_c(t_0) = 0$$

if A, B constant, steady-state value of this DE satisfies

$$\dot{W}_c(t) \to 0 \Rightarrow W_c(t) \to W_{c,ss}$$

$\Rightarrow$ Alternate (#2) characteriz of controllability for Hurwitz (stable) LTI systems:

If soln of Lyap Eq'n

$$AW_{c,ss} + W_{c,ss}A^T = -BB^T$$

has $W_{c,ss}$ positive definite

Matlab "lyap"

LTI controllability test #3:

In modal basis $\underline{x}(t) = \sum_{K=1}^{n} z_K(t)\,\underline{v}_K$

$\{\underline{v}_K\}$ eigenvectors of A

Assuming non-repeated Eigenvalues

$$\dot{z}_K(t) = \lambda_K z_K(t) + \sum_{j=1}^{m} (\underline{\omega}_K^T \underline{b}_j)\, u_j(t)$$

λ_K = eigenvalues of A

$\underline{\omega}_K^T$ = left eigenvectors of A

"modal ctlb. test"

if $\underline{\omega}_K^T \underline{b}_j = 0$ for some K, and all $j = 1, \dots, m$

then $\dot{z}_K(t) = \lambda_K z_K(t) \Rightarrow z_K(t) = z_K(0) e^{\lambda_K t}$

and we cannot change this.

$\Rightarrow$ we cannot create arbitrary changes to $\underline{x}(t)$ along direction $\underline{v}_K$.

$\Rightarrow$ system is not controllable.

recall modal B matrix

$$B_m = V^{-1} B$$

$$= \begin{bmatrix} \underline{\omega}_1^T \\ \underline{\omega}_2^T \\ \vdots \\ \underline{\omega}_n^T \end{bmatrix} B = \begin{bmatrix} \underline{\omega}_1^T B \\ \underline{\omega}_2^T B \\ \vdots \\ \underline{\omega}_n^T B \end{bmatrix}$$

arbitrary (K^{th}) row of B_m looks like

$$\underline{\omega}_K^T B = \begin{bmatrix} (\underline{\omega}_K^T \underline{b}_1) & (\underline{\omega}_K^T \underline{b}_2) & \cdots & (\underline{\omega}_K^T \underline{b}_m) \end{bmatrix}$$

if this cond'n satisfied, then

$$\underline{\omega}_K^T B = \begin{bmatrix} 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$\Rightarrow \underline{\omega}_K^T B = \underline{0}^T$$

If there exists $K = 1, \dots, n$ such that

$$\underline{\omega}_K^T B = \underline{0}^T$$

system is not controllable

"PBH" ctlb test.

LTI modal controllability:

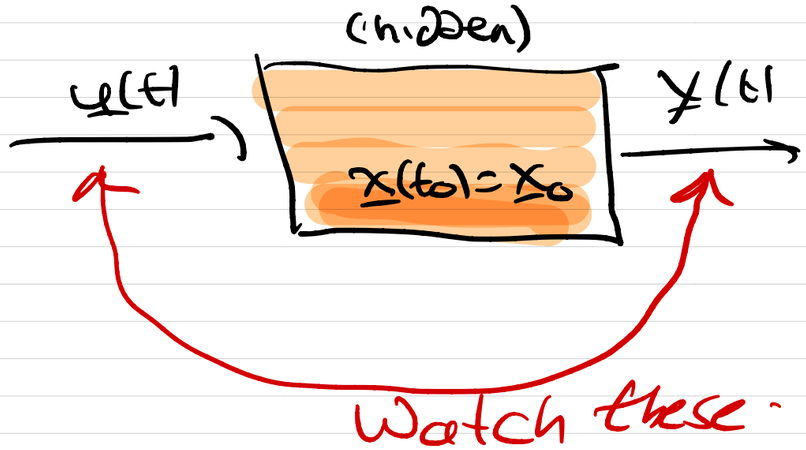
$$\underline{w}_K^T \underline{B} = \underline{0}^T \iff K^{th} \text{ mode uncontrollable}$$

if K^{th} mode uncontrollable, but stable $Re\{\lambda_K\} < 0$, the mode is said to be "stabilizable".

A system is stabilizable if all its uncontrollable modes are stabilizable.

Observability

A system is observable at t_0 if there exists finite $t_1 > t_0$ such that for any input $\underline{u}(t)$, $t \in [t_0, t_1]$ and any IC $\underline{x}(t_0) = \underline{x}_0$, we can calculate (in principle!) $\underline{x}_0$ using measurements $\underline{y}(t)$, $t \in [t_0, t_1]$



We know:

$$\underline{y}(t) = C(t)\underline{x}(t) + D(t)\underline{u}(t)$$

$$\dot{\underline{x}}(t) = A(t)\underline{x}(t) + B(t)\underline{u}(t)$$

$$\Rightarrow \underline{x}(t) = \Phi(t, t_0)\underline{x}_0 + \int_{t_0}^{t} \Phi(t, \tau) B(\tau) \underline{u}(\tau) d\tau$$

Assume: B, C, D Known

$A \leftrightarrow \Phi$ Known

$\underline{y}(t)$, $\underline{u}(t)$ measured.

$$\underline{y}(t) = C(t)\Phi(t,t_0)\underline{x}_0 + \int_{t_0}^{t} C(t)\Phi(t,\tau)B(\tau)\underline{u}(\tau)\,d\tau + D(t)\underline{u}(t)$$

So

$$C(t)\Phi(t,t_0)\underline{x}_0 = \underline{\tilde{y}}(t)$$

$p \times n$ $\quad$ $n \times n$

$p \times n$ $\quad$ p = # of outputs

if $p=n$ and $C(t)$ is invertible for every t, then we can just multiply by inverse to isolate $\underline{x}_0$.

But typically $p \ll n$, try to "square up" problem

$$\Phi^T(t,t_0)C^T(t)C(t)\Phi(t,t_0)\underline{x}_0 = \Phi^T(t,t_0)C^T(t)\underline{\tilde{y}}(t)$$

$n \times n$ matrix

But max rank is only $p<n$.

Not invertible at any t